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Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

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Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

0.1 Note

Treat this document as an interim results repository. Grammar, notation and style may be inconsistent, references may be made to simulations not yet ran, or simulations may be presented without any reference to them throughout the text. The proofs are found in the separate "tablet notes" pdf.

1 Introduction

Foundational papers on Median-of-Means: Nemirovskij and Yudin (1983) and Jerrum, Valiant and Vazirani (1986). For giving the framework and template on the proofs: Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

2 Median-of-Means (MoM) Estimator

The following section is adapted from Devroye et al. (2016) and Lugosi and Mendelson (2019)

2.1 Sub-Gaussian Estimators

Definition 1. *Sub-Gaussian Estimators* Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ . For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if

$$\mathbb{P}\left(|\bar{\mu}(X) - \mu| > L\sigma\sqrt{\frac{\ln(1/\delta)}{n}}\right) \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (1)$$

2.2 The Empirical Mean Estimator

Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$. The empirical mean is defined as

$$\bar{\mu} = \frac{1}{n} \sum_{i=1}^n X_i.$$

By the Central Limit Theorem,

$$\sqrt{n}(\bar{\mu} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

which implies that for any fixed $\delta \in (0, 1)$,

$$\mathbb{P}\left(|\bar{\mu} - \mu| > \frac{\sigma\Phi^{-1}(1 - \delta/2)}{\sqrt{n}}\right) \rightarrow \delta \quad \text{as } n \rightarrow \infty.$$

Using the bound $\Phi^{-1}(1 - \delta/2) \leq \sqrt{2 \ln(2/\delta)}$, it follows that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(|\bar{\mu} - \mu| > \sigma \sqrt{\frac{2 \ln(2/\delta)}{n}} \right) \leq \delta,$$

so $\bar{\mu}$ is an asymptotically sub-Gaussian estimator in the sense of (1) with constant $L = \sqrt{2}$.

While $\bar{\mu}$ is asymptotically sub-Gaussian, it is not a sub-Gaussian estimator for finite samples; Chebyshev's inequality gives

$$\mathbb{P} \left(|\bar{\mu} - \mu| > \sigma \sqrt{\frac{1}{n\delta}} \right) \leq \delta$$

shows that the dependence on δ is polynomial rather than logarithmic, which is substantially worse than (1) for small δ . As such, the empirical mean is insufficient as a sub-Gaussian estimator for random variables with only finite first and second moments.

2.3 The Median-of-Means Estimator

The median-of-means (MoM) estimator is not new, being proposed in various forms, first by Nemirovskij and Yudin (1983) and Jerrum et al. (1986). The algorithm works as follows

Algorithm 1: Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;

return $\hat{\mu} = \text{median}(\bar{\mu}_1, \dots, \bar{\mu}_k)$;

The MoM estimator satisfies

$$\mathbb{P} \left(|\hat{\mu} - \mu| > \sigma \sqrt{\frac{4}{m}} \right) \leq e^{-k/8},$$

and setting $k = \lceil 8 \ln(1/\delta) \rceil$, this becomes

$$\mathbb{P} \left(|\hat{\mu} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}} \right) \leq \delta \quad \text{for all } \delta \in (0, 1) \quad (2)$$

which is sub-Gaussian with $L = \sqrt{32}$

3 Median-of-Means Two-Stage Least Squares (MoM-2SLS)

We first explore the MoM estimator in a univariate setting, with one instrument and one variable.

3.1 Univariate MoM-2SLS

In the univariate setting, the 2SLS estimator collapses to the Wald/IV estimator

$$\hat{\beta}_{IV} = \frac{\sum_{i=1}^n Z_i Y_i}{\sum_{i=1}^n Z_i X_i} = \frac{\bar{S}_{ZY}}{\bar{S}_{ZX}}.$$

This is a ratio of two sample means. To solve this, we can utilise two MoM-type estimators. The first replaces each sample mean with its MoM counterpart; we call this the ratio-of-medians estimator. The second returns the median of the Wald/IV estimate of each block, we call this the median-of-ratios estimator.

3.2 Setup

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i,$$

where $\mathbb{E}[Z_i \varepsilon_i] = 0$ (exogeneity) and $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$ (relevance). Define

$$\mu_{ZY} := \mathbb{E}[Z_i Y_i], \quad \sigma_{ZY}^2 := \text{Var}(Z_i Y_i), \quad \sigma_{ZX}^2 := \text{Var}(Z_i X_i), \quad \sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i),$$

all assumed finite. Both estimators partition the sample into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$ and compute block means

$$\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i, \quad \hat{S}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i.$$

They differ only in how the block-level quantities are aggregated.

3.3 The Estimators

Definition 2 (Ratio-of-Medians (RoM)).

$$\hat{\beta}_{\text{RoM}} = \frac{\tilde{S}_{ZY}}{\tilde{S}_{ZX}}, \quad \text{where } \tilde{S}_{ZY} = \text{median}_j(\hat{S}_{ZY}^{(j)}), \quad \tilde{S}_{ZX} = \text{median}_j(\hat{S}_{ZX}^{(j)}).$$

The medians are taken separately and the ratio is formed afterwards.

Definition 3 (Median-of-Ratios (MoR)).

$$\hat{\beta}_{\text{MoR}} = \text{median}_j(\hat{\beta}^{(j)}), \quad \text{where } \hat{\beta}^{(j)} = \frac{\hat{S}_{ZY}^{(j)}}{\hat{S}_{ZX}^{(j)}}.$$

The ratio is formed within each block and the median is taken over the block-level estimates.

3.4 Concentration Results

Theorem 4 (Standard IV). *Suppose $n \geq 8\sigma_{ZX}^2/(\delta \mu_{ZX}^2)$, then*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}\right] \leq \delta. \quad (3)$$

Theorem 5 (Ratio-of-Medians). *Let $k = \lceil 8 \ln(2/\delta) \rceil$ and $m = \lfloor n/k \rfloor$. Suppose $m \geq 4\sigma_{ZX}^2/\mu_{ZX}^2$. Then*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{RM}} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \quad (4)$$

Theorem 6 (Median-of-Ratios). *Let $k = \lceil 8 \ln(1/\delta) \rceil$ and $m = \lfloor n/k \rfloor$. Suppose $m \geq 32\sigma_{ZX}^2/\mu_{ZX}^2$. Then*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{MR}} - \beta\right| > \frac{4\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}\right] \leq \delta. \quad (5)$$

The proofs for these results can be found in the tablet notes.

Note: This paragraph here is so that I do not forget. The error bounds can then be written as

$$\left|\hat{\beta} - \beta\right| \leq C \cdot \frac{\sigma_{\text{eff}}}{|\mu_{ZX}|} \cdot g(\delta, n),$$

where C is an absolute constant, σ_{eff} is an effective standard deviation, and $g(\delta, n)$ captures the dependence on confidence and sample size. Note that this definition is directly comparable to 1.

3.4.1 Numerical Impact

δ	$1/\sqrt{\delta}$ (IV)	$\sqrt{\ln(2/\delta)}$ (RoM)	$\sqrt{\ln(1/\delta)}$ (MoR)	IV / MoR ratio
0.10	3.16	1.73	1.52	2.1
0.05	4.47	1.92	1.73	2.6
0.01	10.0	2.30	2.15	4.7
10^{-3}	31.6	2.74	2.63	12.0
10^{-6}	10^3	3.80	3.72	269
10^{-9}	3.16×10^4	4.60	4.55	6950

3.4.2 Effective Variance

The RoM and IV bound have $\sigma_{ZY} + |\beta|\sigma_{ZX}$ in the numerator while the MoR has $\sigma_{Z\varepsilon}$. Using $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, it can be shown that

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\sigma_{ZX}.$$

Thus, the MoR and IV bounds are tighter. The proof is found in the tablet notes.

3.4.3 Instrument Strength Condition

	Standard IV	RoM	MoR
Min. block size	-	$\frac{4\sigma_{ZX}^2}{\mu_{ZX}^2}$	$\frac{32\sigma_{ZX}^2}{\mu_{ZX}^2}$
Min. sample size	$\frac{8\sigma_{ZX}^2}{\delta \mu_{ZX}^2}$	$\frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \ln(2/\delta)$	$\frac{256\sigma_{ZX}^2}{\mu_{ZX}^2} \ln(1/\delta)$
δ dependence	Polynomial	Logarithmic	Logarithmic

The standard IV instrument strength condition depends inversely on $1/\delta$, meaning that achieving higher confidence requires proportionally more samples. The MoR and RoM estimators depend logarithmically on δ , which is a significant improvement over the standard IV.

Comparing the two MoM-based estimators, the RoM condition is eight times less stringent than the MoR condition.

3.4.4 Conclusion for the Univariate Case

No single estimator dominates. The standard IV has the tightest constants but the weakest instrument strength for moderate δ . Its polynomial tail makes it unsuitable for high-confidence inference. The MoR achieves the tightest sub-Gaussian bound, but requires the strongest instrument condition. The RoM has a mild instrument condition, but a looser effective variance.

4 Simulation Results

We now present the results of some simulations ran to verify the theoretical results and compare the models

4.1 Data Generating Process

We generate i.i.d. samples $(Y_i, X_i, Z_i)_{i=1}^n$ from

$$Y_i = \beta X_i + \varepsilon_i, \quad X_i = \mu_{ZX} Z_i + v_i,$$

with $\beta = 2$, $\mu_{ZX} = 1$, $\sigma_v = 0.8$, and $Z_i \sim \text{Uniform}\{-1, +1\}$. The errors ε_i, v_i are independent of Z_i and of each other.

Our baseline distribution is $\text{Pareto}(\alpha = 2.1)$, centred with zero mean and finite variance but no finite moments of order $2\alpha = 4.2$, placing it in the heavy-tailed regime where sub-Gaussian concentration matters most. We also consider $\text{Student-}t(3)$, $\text{Pareto}(\alpha \in \{2.5, 3.5\})$, and Gaussian errors. Unless stated otherwise, the $\text{Pareto}(\alpha = 2.1)$ is assumed to be the baseline distribution.

Population quantities $\sigma_{Z\varepsilon} \approx 2.88$, $\sigma_{ZX} \approx 2.57$, $\mu_{ZX} \approx 1.00$ are estimated from a single sample of $n = 500,000$ and used for theoretical bounds throughout. Unless stated otherwise, δ is set to 0.05.

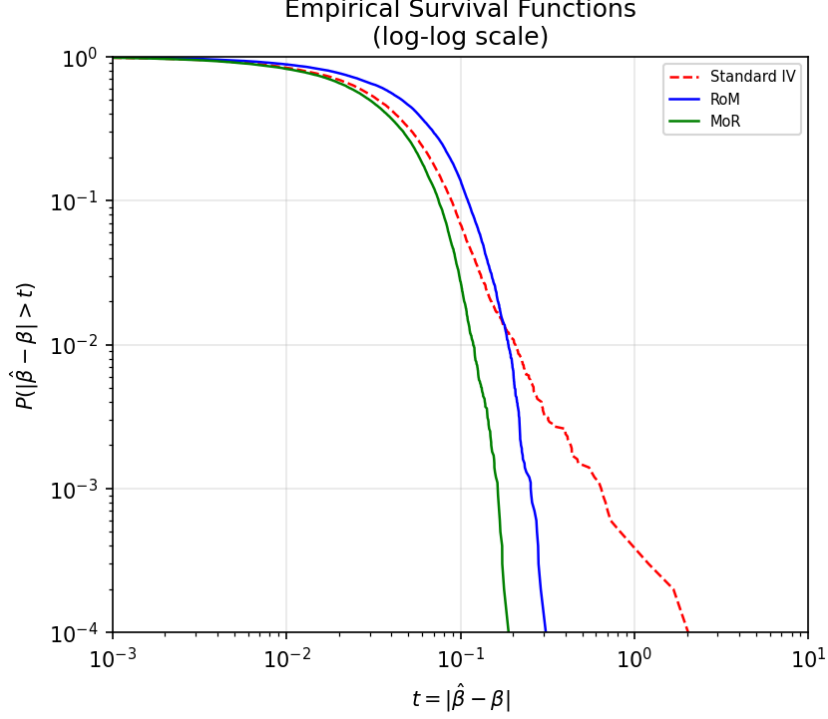


Figure 1: Empirical survival functions $\hat{P}(|\hat{\beta} - \beta| > t)$ on a log-log scale ($n = 2,000$, $k = 16$, Pareto(2.1) errors, 10,000 replications).

4.2 Empirical Survival Functions

We draw 10,000 independent samples of size $n = 2,000$ under Pareto(2.1) errors and plot the empirical survival function $\hat{P}(|\hat{\beta} - \beta| > t)$ for each estimator on a log-log scale (Figure 1). For sufficiently small δ , standard IV becomes the worst performing. MoR shows the steepest decay, consistent with its tighter bound found earlier.

4.3 Bound Tightness and Coverage

We assess conservativeness at failure probabilities $\delta \in \{0.01, 0.05, 0.10, 0.30, 0.50\}$. For each δ we set $k = \lceil 8 \ln(1/\delta) \rceil$ (MoR) or $k = \lceil 8 \ln(2/\delta) \rceil$ (RoM), compute the theoretical CI width, and measure empirical coverage over 10,000 replications with $n = 2,000$. We report c^* , the smallest fraction of the theoretical width achieving coverage $\geq 1 - \delta$; $1/c^*$ measures conservativeness.

Table 1 shows all bounds achieve nominal coverage. The IV bound is ≈ 6 – $10\times$ conservative, MoR is ≈ 21 – $28\times$, and RoM is the most conservative at ≈ 50 – $70\times$.

4.3.1 Theoretical Widths at Extreme δ

Table 2 reports theoretical CI widths for $\delta \in [10^{-1}, 10^{-8}]$ without simulation. The IV width grows as $1/\sqrt{\delta}$ while MoR grows as $\sqrt{\ln(1/\delta)}$, with a crossover near $\delta \approx 5.9 \times 10^{-3}$. At $\delta = 10^{-6}$ the IV bound is $47\times$ wider, confirming the qualitative advantage of sub-Gaussian concentration for high-confidence inference.

Table 1: Coverage validation at moderate δ . c^* is the smallest fraction of the theoretical half-width needed to achieve coverage $\geq 1 - \delta$; $1/c^*$ measures conservativeness.

δ	k_{MoR}	m	Theoretical width			Conservativeness ($1/c^*$)		
			IV	MoR	RoM	IV	MoR	RoM
0.50	6	333	0.26	0.89	2.84	$7.8\times$	$27.8\times$	$62.5\times$
0.30	10	200	0.33	1.15	3.64	$6.4\times$	$24.4\times$	$52.6\times$
0.10	19	105	0.58	1.59	5.53	$6.6\times$	$21.7\times$	$50.0\times$
0.05	24	83	0.81	1.79	7.35	$7.3\times$	$21.3\times$	$55.6\times$
0.01	37	54	1.82	2.21	13.32	$9.9\times$	$21.3\times$	$71.4\times$

Table 2: Theoretical CI widths at extreme δ . MoR becomes tighter than IV for $\delta \lesssim 5.9 \times 10^{-3}$.

δ	IV width	MoR width	IV/MoR ratio	k_{MoR}	m
10^{-1}	0.58	1.59	0.36	19	105
10^{-2}	1.82	2.21	0.82	37	54
10^{-3}	5.75	2.75	2.09	56	35
10^{-4}	18.19	3.13	5.81	74	27
10^{-6}	181.92	3.84	47.4	111	18
10^{-8}	1819.2	4.51	403	148	13

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